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In [1]: import tensorflow as tf
import matplotlib.pyplot as plt

from tensorflow.keras.datasets import mnist
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense
from tensorflow.keras.utils import to_categorical

(x_train, y_train), (x_test, y_test) = mnist.load_data()

x_train = x_train.astype("float32") / 255.0
x_test = x_test.astype("float32") / 255.0

x_train = x_train.reshape(-1, 28, 28, 1)
x_test = x_test.reshape(-1, 28, 28, 1)

y_train = to_categorical(y_train, 10)
y_test = to_categorical(y_test, 10)

model1 = Sequential([
    Conv2D(
        32,
        (3, 3),
        activation="relu",
        input_shape=(28, 28, 1)
    ),
    MaxPooling2D((2, 2)),
    Flatten(),
    Dense(
        128,
        activation="relu"
    ),
    Dense(
        10,
        activation="softmax"
    )
])

model1.compile(
    optimizer="adam",
    loss="categorical_crossentropy",
    metrics=["accuracy"]
)
```

```
model1.fit(
    x_train,
    y_train,
    epochs=5,
    batch_size=64,
    verbose=0
)

loss1, accuracy1 = model1.evaluate(
    x_test,
    y_test,
    verbose=0
)

model2 = Sequential([
    Conv2D(
        32,
        (3, 3),
        activation="relu",
        input_shape=(28, 28, 1)
    ),

    MaxPooling2D((2, 2)),

    Conv2D(
        64,
        (3, 3),
        activation="relu"
    ),

    MaxPooling2D((2, 2)),

    Flatten(),

    Dense(
        128,
        activation="relu"
    ),

    Dense(
        10,
        activation="softmax"
    )
])

model2.compile(
    optimizer="adam",
    loss="categorical_crossentropy",
    metrics=["accuracy"]
)
```

```
model2.fit(
    x_train,
    y_train,
    epochs=5,
    batch_size=64,
    verbose=0
)

loss2, accuracy2 = model2.evaluate(
    x_test,
    y_test,
    verbose=0
)

pred1 = model1.predict(
    x_test[:2],
    verbose=0
)

pred2 = model2.predict(
    x_test[:2],
    verbose=0
)

print("\n=====")
print("MODEL 1 RESULTS")
print("=====")

print(
    "Accuracy:",
    round(accuracy1 * 100, 2),
    "%"
)

print(
    "Loss:",
    round(loss1, 4)
)

plt.figure(figsize=(8, 4))

for i in range(2):
    predicted = pred1[i].argmax()
```

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actual = y_test[i].argmax()

vector = y_test[i].astype(int)

vector_text = "[" + ",".join(
    map(str, vector)
) + "]"

plt.subplot(
    1,
    2,
    i + 1
)

plt.imshow(
    x_test[i].reshape(28, 28),
    cmap="gray"
)

plt.title(
    f"Predicted: {predicted}\n"
    f"Actual: {actual}\n"
    f"One-Hot: {vector_text}"
)

plt.axis("off")

plt.suptitle(
    "Model 1 Predictions"
)

plt.tight_layout()

plt.show()

print("\n=====")
print("MODEL 2 RESULTS")
print("=====")

print(
    "Accuracy:",
    round(accuracy2 * 100, 2),
    "%"
)

print(
    "Loss:",
    round(loss2, 4)
)

plt.figure(figsize=(8, 4))

for i in range(2):

```

```
predicted = pred2[i].argmax()

actual = y_test[i].argmax()

vector = y_test[i].astype(int)

vector_text = "[" + ",".join(
    map(str, vector)
) + "]"

plt.subplot(
    1,
    2,
    i + 1
)

plt.imshow(
    x_test[i].reshape(28, 28),
    cmap="gray"
)

plt.title(
    f"Predicted: {predicted}\n"
    f"Actual: {actual}\n"
    f"One-Hot: {vector_text}"
)

plt.axis("off")

plt.suptitle(
    "Model 2 Predictions"
)

plt.tight_layout()

plt.show()
```

WARNING:tensorflow:From C:\Users\Student\AppData\Roaming\Python\Python310\site-packages\keras\src\losses.py:2976: The name tf.losses.sparse\_softmax\_cross\_entropy is deprecated. Please use tf.compat.v1.losses.sparse\_softmax\_cross\_entropy instead.

WARNING:tensorflow:From C:\Users\Student\AppData\Roaming\Python\Python310\site-packages\keras\src\backend.py:873: The name tf.get\_default\_graph is deprecated. Please use tf.compat.v1.get\_default\_graph instead.

WARNING:tensorflow:From C:\Users\Student\AppData\Roaming\Python\Python310\site-packages\keras\src\layers\pooling\max\_pooling2d.py:161: The name tf.nn.max\_pool is deprecated. Please use tf.nn.max\_pool2d instead.

WARNING:tensorflow:From C:\Users\Student\AppData\Roaming\Python\Python310\site-packages\keras\src\optimizers\\_\_init\_\_.py:309: The name tf.train.Optimizer is deprecated. Please use tf.compat.v1.train.Optimizer instead.

WARNING:tensorflow:From C:\Users\Student\AppData\Roaming\Python\Python310\site-packages\keras\src\utils\tf\_utils.py:492: The name tf.ragged.RaggedTensorValue is deprecated. Please use tf.compat.v1.ragged.RaggedTensorValue instead.

WARNING:tensorflow:From C:\Users\Student\AppData\Roaming\Python\Python310\site-packages\keras\src\engine\base\_layer\_utils.py:384: The name tf.executing\_eagerly\_outside\_functions is deprecated. Please use tf.compat.v1.executing\_eagerly\_outside\_functions instead.

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#### MODEL 1 RESULTS

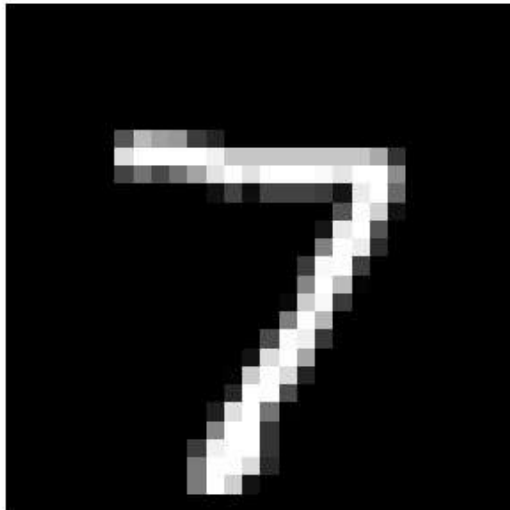
=====

Accuracy: 98.7 %

Loss: 0.0442

#### Model 1 Predictions

Predicted: 7  
Actual: 7  
One-Hot: [0,0,0,0,0,0,0,1,0,0]



Predicted: 2  
Actual: 2  
One-Hot: [0,0,1,0,0,0,0,0,0,0]



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#### MODEL 2 RESULTS

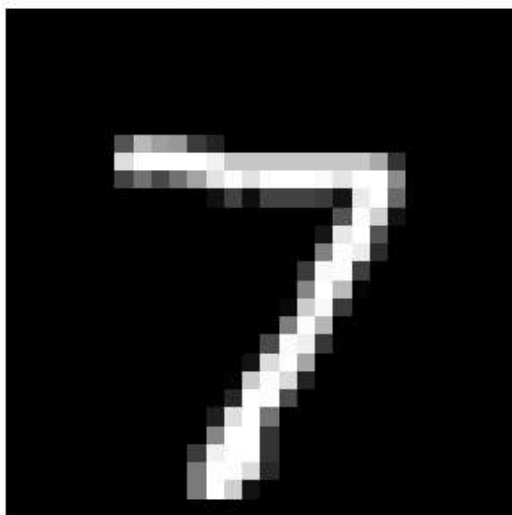
=====

Accuracy: 98.88 %

Loss: 0.033

## Model 2 Predictions

Predicted: 7  
Actual: 7  
One-Hot: [0,0,0,0,0,0,0,1,0,0]



Predicted: 2  
Actual: 2  
One-Hot: [0,0,1,0,0,0,0,0,0,0]



In [ ]: